National University of Computer & Emerging Sciences
MT-2005 Probability and Statistics
Spring 2026

Course Content for Final Exam

<table>
<tr><th>Contents/Topics</th><th>Question set</th><th>Marks distribution</th></tr>
<tr><td>Graphical representation of data:
Construction of bar chart, histograms, Stem-leaf plots.
Sample Space and Event:
Tree diagram, Counting techniques</td><td rowspan="9">NW
Exercise # 2.2
Questions 2.18-2.2,2.52,2.53,2.109
Exercise # 3.3
Questions 3.113-
3.135
Questions 3.62-3.78
WP
Questions
2.1-2.13
2.21-2.48.2.51,2.55,2.58
2.49-2.65
2.73-2.89
2.95-2.102
MB
Questions
3.1-3.24, 3.28
3.1-3.24, 3.28
WP
Questions
5.4-5.20, 5.25-5.28,
5.56-5.58,5.67,5.71
MB
Questions
4.1-4.5,4.16-4.26
WP
Examples
6.1-6.8
Questions
6.3-6.9

WP
Questions
9.1-9.5, 9.9- 9.13 and
all solved examples
related to the topic
WP
Questions
10.19-10.46
WP
Questions
11.1-11.9,
11.11-11.14,11.5(a, b),
11.43,11.44,11.46
Examples 11.2-11.5
Questions 11.17-11.22
Questions 13.1-13.7
NW
Questions
16.42-16.47</td><td rowspan="5">30%</td></tr>
<tr><td>Rules of Probability:
Probability of an event, Additive rules, Multiplication Rule, Conditional Probability, Bayes’ Rules</td></tr>
<tr><td>Random Variables
Concept of random variable and their types

Discrete Random Variables and their distributions:
Distribution of random variables:
Main concept of random variables (CLO-1), PMF and CDF and Distribution of random vector:
Joint and Marginal distributions and independence of random variables.</td></tr>
<tr><td>Continuous Probability Distributions
PDF and CDF, Joint Probability Distribution, marginal distribution, Conditional distribution (Bivariate)</td></tr>
<tr><td>Types of random variables:
Binomial, Poisson, Normal and standard normal distributions and applications</td></tr>
<tr><td>Estimation:
Introduction, confidence interval estimation using z & t distributions for single mean and difference between two means</td><td rowspan="4">70%</td></tr>
<tr><td>Hypothesis Testing:
Testing of hypothesis for single mean and difference between two means using z-test and t-test Z(CLO-3), p-value method (CLO-3)</td></tr>
<tr><td>Regression & Correlation:
Scattered plot/diagram.
Correlation coefficient, Introduction to simple linear regression, Relationship between correlation and regression, determining regression equation (the least square method) and Gradient Descent method w.r.t regression (material
Will provide) (CLO-3), Inferences in regression coefficient, Inferences in correlation coefficient, Coefficient of determination.</td></tr>
<tr><td>Analysis of variance:
Introduction to ANOVA , F-test, one-way ANOVA</td></tr>
</table>